

Decentralized Exchanges (DEX) have emerged recently with Uniswap as the leading automated market maker (AMM) by volume [1]. An automated market maker (AMM) is a protocol which permits the execution of buy and sell orders on a blockchain. These protocols play a similar role as limit order books in traditional finance; users deposit their tokens into a liquidity pool, where they can be used by other actors for trading activities. A liquidity provider (LP) is an actor who transfers tokens into liquidity pools and is rewarded by the fees paid by traders who use that pool to swap tokens. As a result, the price moves between so-called “unitary price ranges”, entering and exiting liquidity pools listed for the same exchange pair.

Uniswap v3 is a constant product market maker (CPMM), meaning that it maintains invariant function  $\mathcal{I}(x, y) = x \cdot y = L^2$  where  $L$  denotes the liquidity and  $x$  and  $y$  are virtual reserves of tokens  $\mathcal{X}$  and  $\mathcal{Y}$ , respectively. A virtual reserve for a token  $\mathcal{X}$  is defined as  $x = x_{\text{offset}} + x_r$  where  $x_r$  is the real reserves, and  $x_{\text{offset}}$  are some other quantities. LPs wishing to provide liquidity must specify a price range over which their tokens will be used for trading activities (whilst the price remains in that range). Let square-root price range  $R := [\pi_l, \pi_u]$  where  $\pi_l = \sqrt{p_l} = \sqrt{\frac{x_l}{y_l}}$  and  $\pi_u = \sqrt{p_u} = \sqrt{\frac{x_u}{y_u}}$  such that price  $p(\tau) = (\beta^{(p)})^\tau$  where  $\beta^{(p)} = 1.0001$  and “tick”  $\tau$  is a multiple of  $\delta_\pi \in \mathbb{N}$ . Then, a *unitary* price range  $R_i$  is the range spanned by two consecutive ticks, namely  $R_i = [\pi_{l_i}, \pi_{u_i}]$  such that  $\pi_{l_i}^2 = p(\delta_\pi \cdot i)$  and  $\pi_{u_i}^2 = p(\delta_\pi \cdot (i + 1))$ . Note that the constant-product rule applies here:  $x_u \cdot y_u = x_l \cdot y_l = L^2$ , so each a point  $(x, y)$  must reside on the liquidity curve (see fig. 1), however, in practice the curve is not smooth because liquidity is constant over consecutive unitary price ranges (see fig. 2)

Consider a square-root unitary price  $R = [\pi_l, \pi_u] = R_1 \cup \dots \cup R_n$ , where  $R_i = [\pi_{l_i}, \pi_{u_i}]$  already contains liquidity  $L_i$  for  $i = 1, \dots, n$ . Suppose an LP deposits (resp. withdraws) the same amount of liquidity  $\Delta L$  on range  $R_i$ , such that  $L_i + \Delta L \geq 0 \ \forall i \in \{1, \dots, n\}$ , then he will deposit (resp. withdraw) the following amounts of tokens  $\mathcal{X}$  and  $\mathcal{Y}$  into the pool:

$$\Delta x_r = \Delta L \cdot \left( \frac{1}{\pi_0^R} - \frac{1}{\pi_u} \right) \quad \text{and} \quad \Delta y_r = \Delta L \cdot (\pi_0^R - \pi_l) \quad \text{s.t.} \quad \pi^R = \begin{cases} \pi_u & \text{if } \pi_u < \pi \\ \pi_l & \text{if } \pi < \pi_l \\ \pi & \text{otherwise} \end{cases} \quad (1)$$

If an LP wishes to withdraw liquidity  $\Delta L$  on price range  $R$ , then he would retrieve  $\Delta x_r$  and  $\Delta y_r$  real reserve tokens as specified above, and in addition, earn  $\Delta x_{\text{fee}} = \Delta L \cdot (\Phi_R^{\mathcal{X}}(t) - \Phi_R^{\mathcal{X}}(t_0))$  and  $\Delta y_{\text{fee}} = \Delta L \cdot (\Phi_R^{\mathcal{Y}}(t) - \Phi_R^{\mathcal{Y}}(t_0))$ , where accumulators  $\Phi_R^{\mathcal{X}}$  and  $\Phi_R^{\mathcal{Y}}$  are updated every time a trader swaps tokens while the square-root price is in range  $R$ . More specifically, the accumulators are updated as follows. Suppose  $a_i$  tokens  $\mathcal{X}$  are transferred into the pool in exchange for tokens  $\mathcal{Y}$  via a swap operation, then  $\Phi_R^{\mathcal{X}} \leftarrow \Phi_R^{\mathcal{X}} + \frac{a_i}{1-\phi} \cdot \frac{\phi}{L}$ , where  $L$  is the available liquidity on range  $R$ .  $\frac{a_i}{1-\phi}$  represents the input tokens actually used in the swap, which implies that  $\frac{a_i \phi}{1-\phi}$  represents the accumulated fees from that operation and that  $\frac{a_i}{1-\phi} \cdot \frac{\phi}{L}$  represents the amount of accumulated fees per unit of liquidity on the range  $R$ .

## Impermanent Loss

A liquidity provider may provide liquidity and simultaneously hold tokens outside of the pool. If they withdraws their tokens at a time where the pool price is significantly different from the one at deposit time, the value of their tokens net of fees will be lower than the value of their original tokens if they had not been deposited into the pool. This loss is only materialized when LPs withdraw their tokens from the pool (“burn” operation), which is called *Impermanent Loss* (IL) [5].

Consider two strategies: (i) a liquidity providing strategy where tokens are deposited into a pool, and (ii) a strategy where tokens are withheld from the pool under  $\mathbf{H}_0$  : Tokens  $\mathcal{X}$  and  $\mathcal{Y}$  outside the pool can be valued using pool price (in practice  $\mathbf{H}_0$  may not hold the case due to discrepancies in centralized exchange price listings see fig. 3). The impermanent loss in reference numéraire token  $\mathcal{Y}$  (which would be realized if the LP were to burn his tokens) is given by

$$V_P(t_1) - V_H(t_1) = -\Delta L \cdot (\pi_0^R - \pi_1^R) \cdot \left(1 - \frac{\pi_1^2}{\pi_0^R \cdot \pi_1^R}\right) \quad (2)$$

These results above can be generalized to an arbitrary liquidity curve  $\Delta L_\pi := \Delta L_i$  for  $\pi \in [\pi_{l_i}, \pi_{u_i})$ . Then the function  $\pi \mapsto \Delta L_\pi$  is piecewise constant and right-continuous with a left limit, allowing for integration w.r.t  $\pi$ . Consider a liquidity provider depositing a liquidity curve  $(\Delta L_\pi)_\pi$  to the pool while keeping  $x_0$  tokens  $\mathcal{X}$  and  $y_0$  tokens  $\mathcal{Y}$  outside the pool (under  $\mathbf{H}_0$ ). Assuming the integrability condition

$$\int_1^{+\infty} \frac{\Delta L_\pi}{\pi^2} d\pi + \int_0^1 \Delta L_\pi d\pi < \infty$$

At time  $t_j \in [0, T) : j \in \{0, 1, \dots, N\}$ , when the square-root price is  $\pi_{t_j} = \sqrt{p_{t_j}}$ , the tokens’ value (in numéraire  $\mathcal{Y}$ ) net of swap fees is given by:

$$V_P(t_j) = x_0 \cdot p_{t_j} + y_0 + p_{t_j} \cdot \int_{\pi_{t_j}}^{+\infty} \frac{\Delta L_\pi}{\pi^2} d\pi + \int_0^{\pi_{t_j}} \Delta L_\pi d\pi$$

This representation lets us calculate the so-called “Greeks”: (Delta)  $\frac{\partial V_P(t_0)}{\partial p_0}$  and (Gamma)  $\frac{\partial^2 V_P(t_0)}{\partial p_0^2}$  which gives the sensitivity of the strategy’s global position  $V_P(t_0)$  with respect to the  $X - Y$  exchange rate.

## Fees Modeling

Token swaps are assumed to occur because of arbitrageurs between the pool and another reference exchange (due to pricing discrepancies, see fig. 3)[2]. Let latent price process  $(p_t)_t$  be defined on a filtered probability space  $(\Omega, \mathcal{F}, (\mathcal{F})_{0 \leq t \leq T}, \mathbb{P})$ . Let hypothesis  $\mathbf{H}_p$  : the latent price process follows an Itô dynamics of the form:  $dp_t/p_t = \mu_t dt + \sigma_t dW_t$  with progressively-measurable drift  $(\mu_t)_t$  and volatility  $(\sigma_t)_t$ : both  $\mu$  and  $\sigma$  are bounded. Swap trades are assumed to occur at discrete swap times  $(t_i)_i : t_{i+1} = \inf\{t \geq t_i : p_t = p_{t_i}(1+\gamma)^{\pm 1}\}$  i.e. when  $(p_t)$  changes by  $\gamma = \phi \in \{0.01\%, 0.03\%, 0.05\%\}$  where  $\phi$  is the swap fee paid out to liquidity providers. This hypothesis allows for the creation of

a risk-neutral probability measure under the filtration via the Girsanov transform [7]

$$\mathbb{P}^*|_{\mathcal{F}_t} = \exp\left(-\int_0^T \frac{\mu_s}{\sigma_s} dW_s - \frac{1}{2} \int_0^T \left(\frac{\mu_s}{\sigma_s}\right)^2 ds\right) \quad (3)$$

Consider an LP depositing a liquidity curve  $(\Delta L_\pi)_\pi$  at time  $t_0$ . Suppose also that the risk-neutral probability rule holds for Calls and Puts (strike  $b$ , expiry  $T$ ) written on the X-Y exchange rate  $\text{Call}_{t=0}(T, b) = \mathbb{E}^*[(p_T - b)_+]$  and  $\text{Put}_{t=0}(T, b) = \mathbb{E}^*[(b - p_T)_+]$ . Then, the amount of total fees  $\mathcal{F} \equiv \mathcal{F}((\Delta L_\pi)_\pi, (\sigma_t)_t)$  accumulated over the period  $[0, T]$  is given by:

$$\mathcal{F} = \int_0^{+\infty} (\Delta L_{\sqrt{b}}) \frac{\mathbb{E}^*[A_T^b(p)]}{2 \cdot b^{3/2}} db = \int_0^{p_0} (\Delta L_{\sqrt{b}}) \cdot \frac{\text{Put}_{t=0}(T, b)}{b^{3/2}} + \int_{p_0}^{+\infty} (\Delta L_{\sqrt{b}}) \cdot \frac{\text{Call}_{t=0}(T, b)}{b^{3/2}} \quad (4)$$

where  $A_T^b$  is the local time of the price process  $p$  at level  $a$  i.e. how much time  $p_t$  spends in price range  $[b, b + \epsilon)$  over time range  $[0, T]$ . Exact fee calculations for a simulated price process, show an increase in fees for both tokens coincide with high local times (see fig. 4)

## Conclusion & Further Research

In this paper, M.Echenim et. al. seek to precisely define the way in which the Uniswap v3 protocol behaves to create an unambiguous knowledge-base for researchers and practitioners. Fan et. al. provides an expression for an LP's profit and loss in v2 and v3 protocols, however their resulting expression was dependent on the path price itself [6]; M.Echenim et. al. remove the path-dependancy and provide a closed-form approximation in eq. (2). Similarly, Bichuch and Feinstein's paper assume a geometric random walk for the price process [3]. M.Echenim et. al. relaxes these assumptions, providing a general asymptotic formula for fees for an arbitrary continuous price process in eq. (4). Overall, this paper represents a substantial step in the quantitative modeling Uniswap AMMs.

However, since a lot of time is spent developing theory starting from the protocol open-source code itself, the authors leave several investigations with real data open. They encourage that further research: (i) compare fees predicted by equation (4) with those actually observed in the pool, and (ii) perform a quantitative study of arbitrage opportunities between calls/puts and Uniswap v3 pools. Suggestion (i) is challenging because estimating the drift  $\mu_t$  and diffusion  $\sigma_t$  of the Itô price process  $p_t$  from historical data is a difficult inverse problem (see eq. (4)). Similarly, suggestion (ii) requires a conversion from the risk-neutral measure to the historical measure given by the Girsanov transform (see eq. (3)), which also necessitates estimating the drift and diffusion coefficients.

Nevertheless, a qualitative analysis of a sample of historical data [4] reveals a higher concentration of liquidity above  $p_T$  and lower fees per unit liquidity below  $p_T$  (see fig. 5). This variability in the distribution of fees and liquidity across price could be indicative of a bearish sentiment (see price in fig. 6) as most LPs are choosing price ranges above the current price to add liquidity. Modeling such variations in liquidity clustering at different unitary price ranges could lead to more accurate forecasts of swap fees distributions, and optimal liquidity curve placement to reduce impermanent loss.

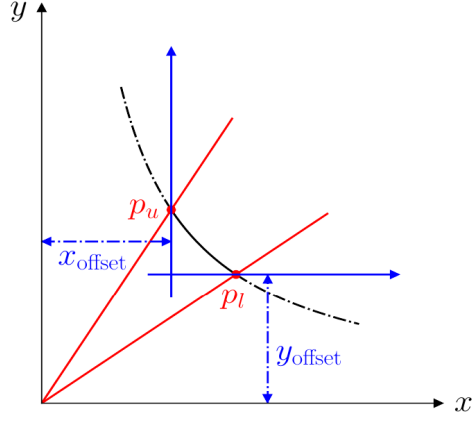


Figure 1: Representation of the virtual numbers  $(x, y)$  and offset numbers  $(x_{\text{offset}}, y_{\text{offset}})$  of tokens [figure from [5]]

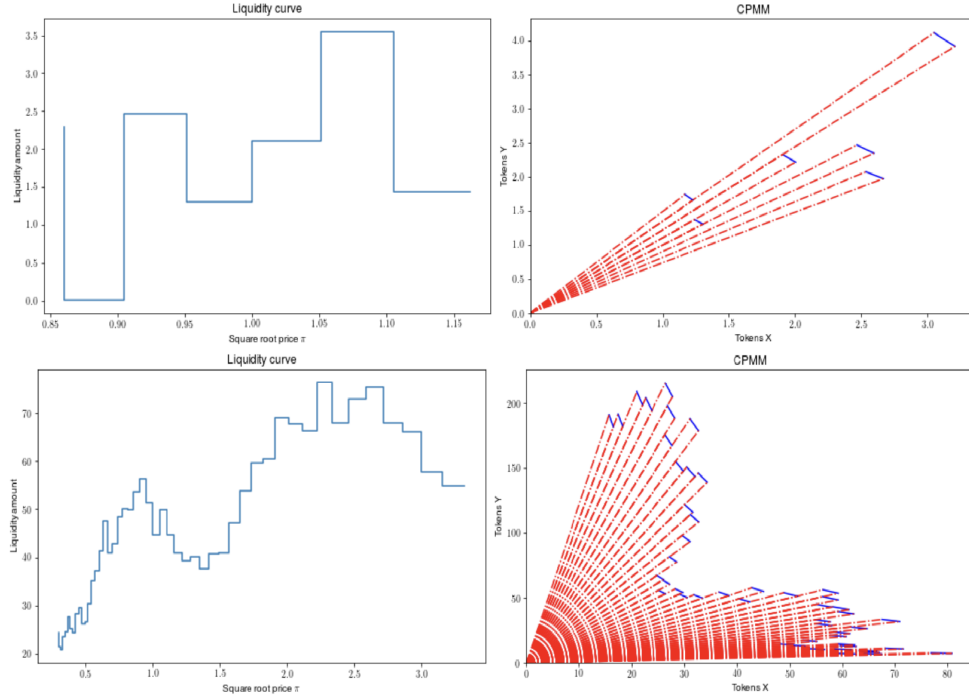


Figure 2: Liquidity quantities deposited over consecutive unitary square-root price ranges (left), and the representation of the CPMM invariant formula  $\mathcal{I}(x, y)$  for each range. [figure from [5]]

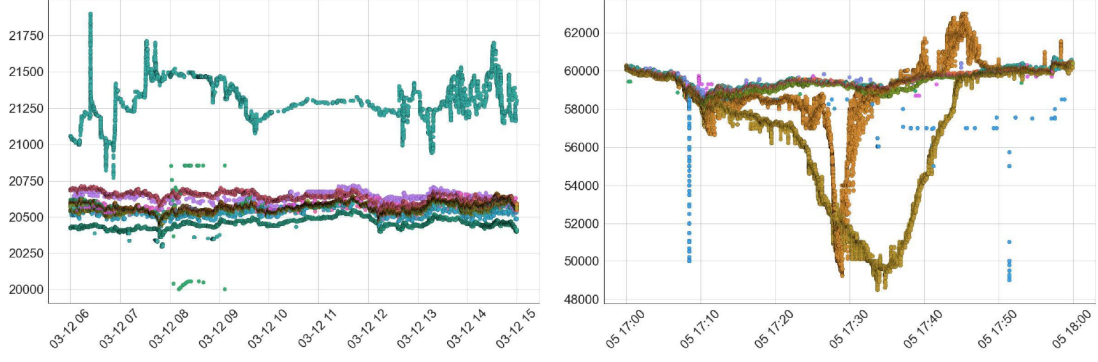


Figure 3: Centralized Exchange (CEX) prices differ from one another with each color represents a different exchange. The trade prices are for the BTC-USD exchange rate on March 12th 2023 (left) and BTC-EUR exchange rate on March 5th 2024 (right)) [figure from [5]]

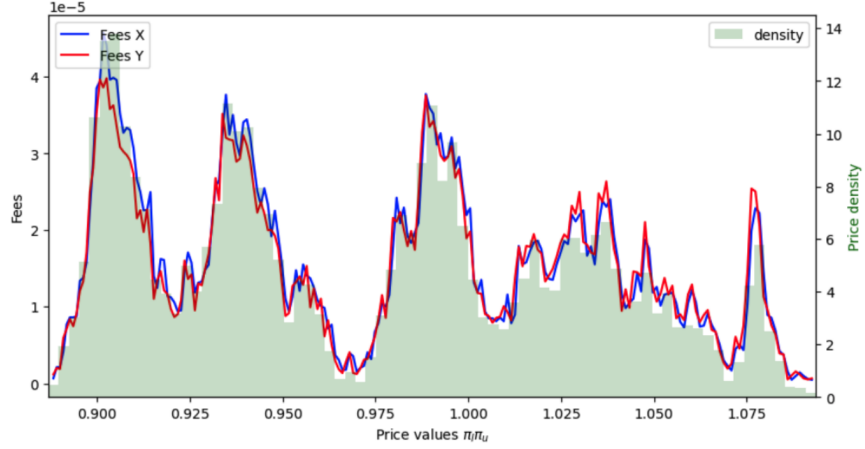


Figure 4: Left axis: exact fees collected in tokens  $\mathcal{X}$  and  $\mathcal{Y}$ . Right axis: occupation density of price process  $(p_t)$  across times  $t \in [0, T]$  where  $T = 1$  month, swap fees  $\gamma = \phi = 0.05\%$ , tick-spacing  $\delta_\pi = 10$  [figure from [5]]

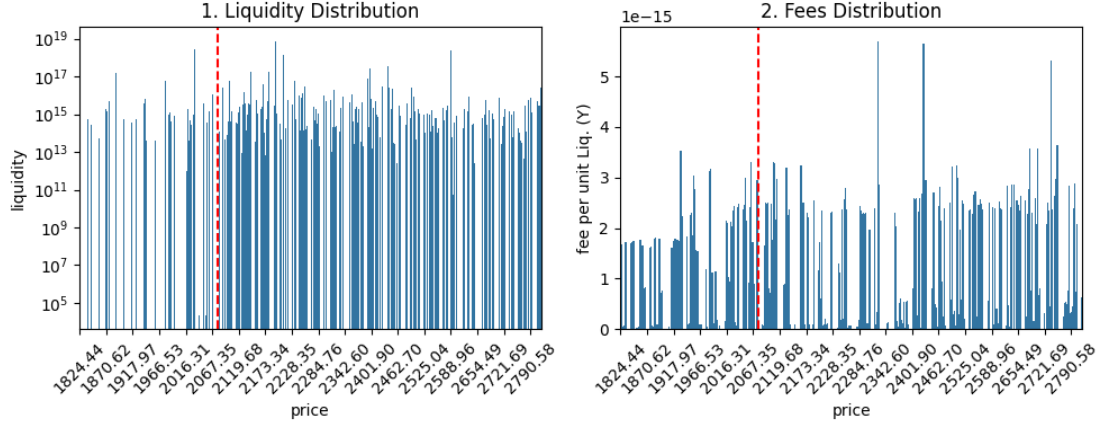


Figure 5: Liquidity distribution and distribution of fees per unit-liquidity  $\Phi_R^{\mathcal{Y}}$  over consecutive unitary price ranges for Wrapped Ethereum WETH (token  $\mathcal{Y}$ ) USDC (token  $\mathcal{X}$ ) for time period Jan. 28th, 2026 to  $T$  = Mar. 5th 2026 (price  $p_T$  indicated by red-dashed line). [figure generated by me: see code [4]]

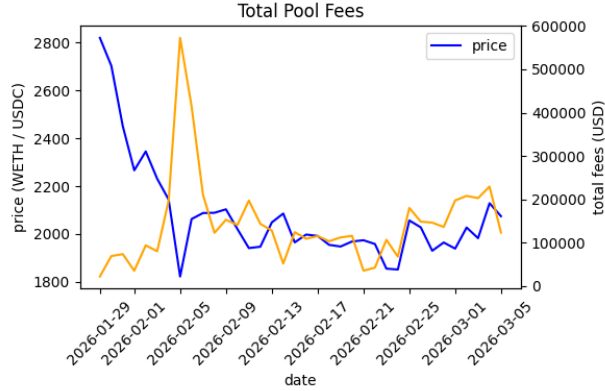


Figure 6: Total liquidity pool fees accumulation from swap transactions between Wrapped Ethereum WETH (token  $\mathcal{Y}$ ) USDC (token  $\mathcal{X}$ ), from Jan. 28th, 2026 to Mar. 5th 2026. [figure generated by me: see code [4]]

## References

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